Polyline Editor – Design & Prototype 1

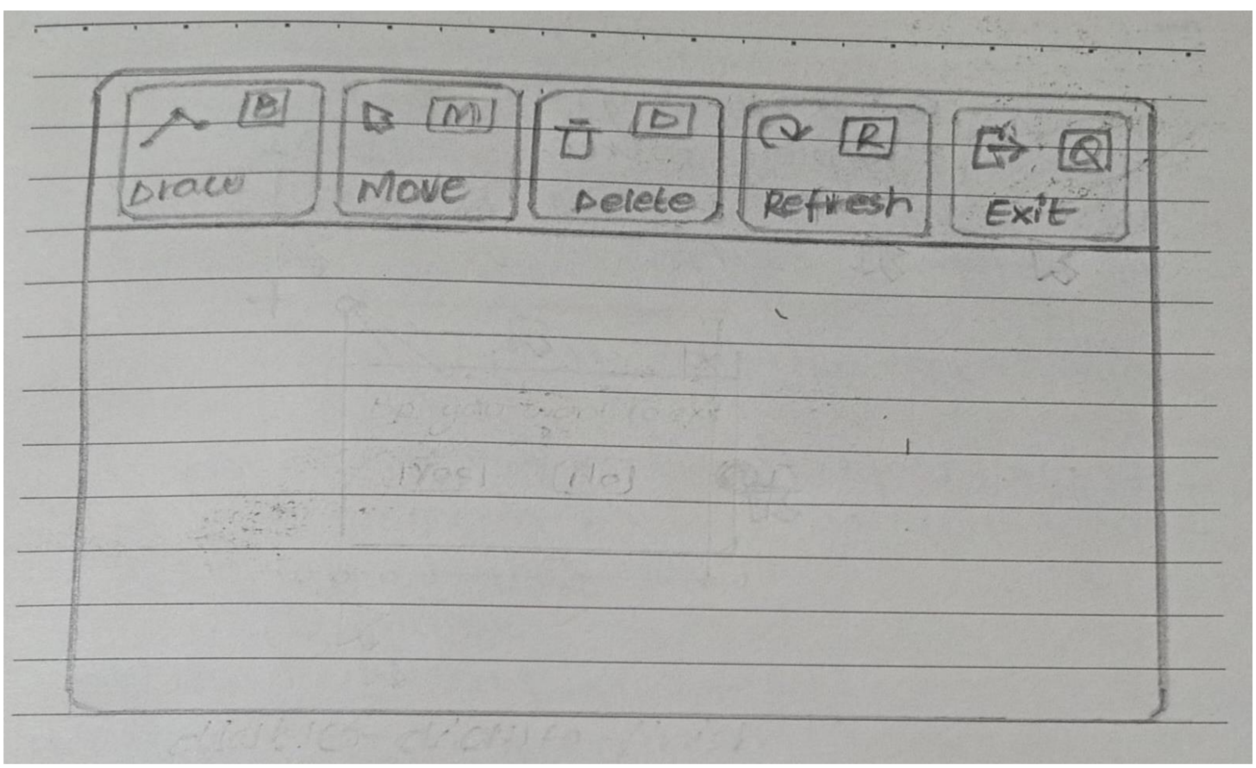
1. Introduction

This document presents the initial design and low-fidelity prototype (Version 1) of the Polyline Editor. The aim is to create a simple and user-friendly system for drawing and editing polylines using basic keyboard and mouse interactions.

2. Design Goals

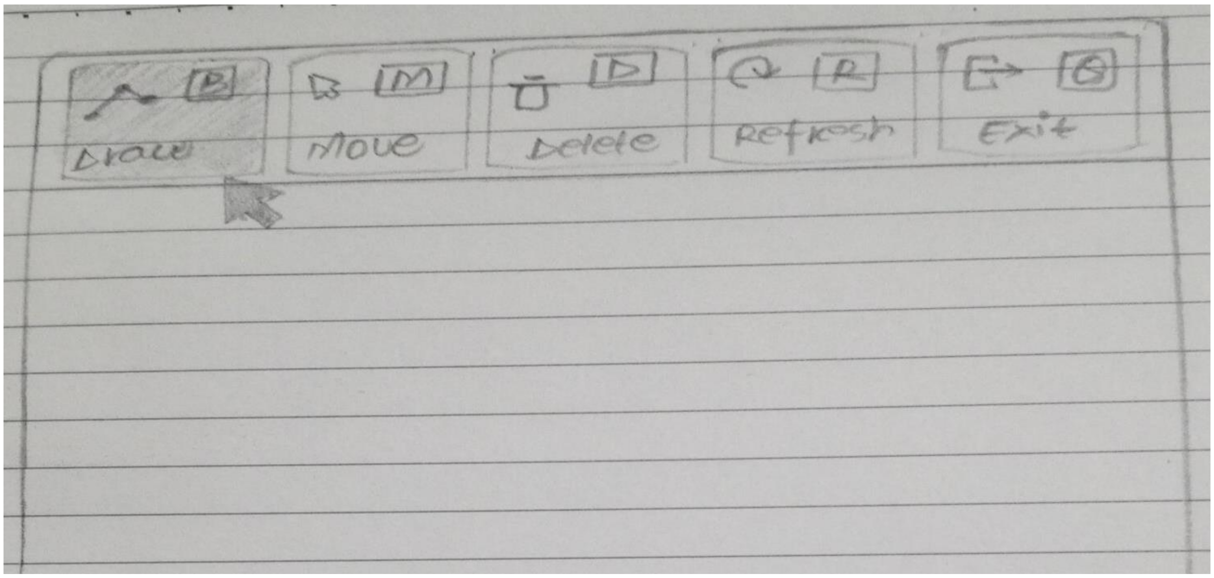
- Simple and intuitive interaction
- Fast and efficient editing of polylines
- Clear system feedback for user actions
- Support error prevention and easy correction
- Maintain consistency in controls and actions

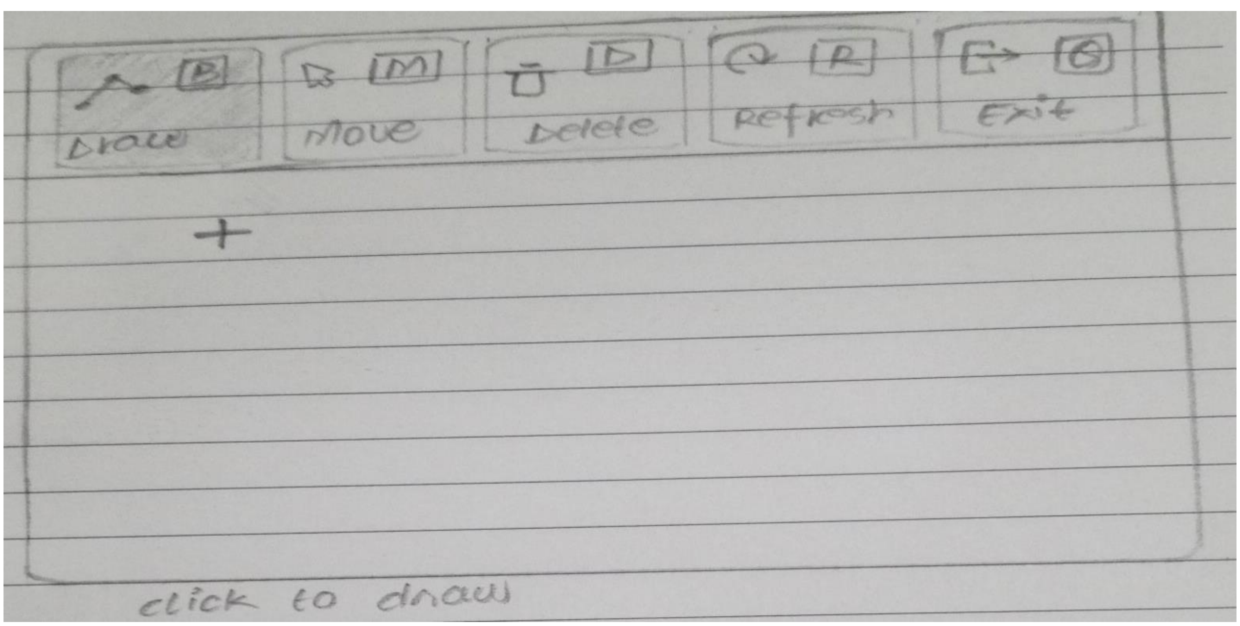
3. Low-Fidelity Interface Sketch

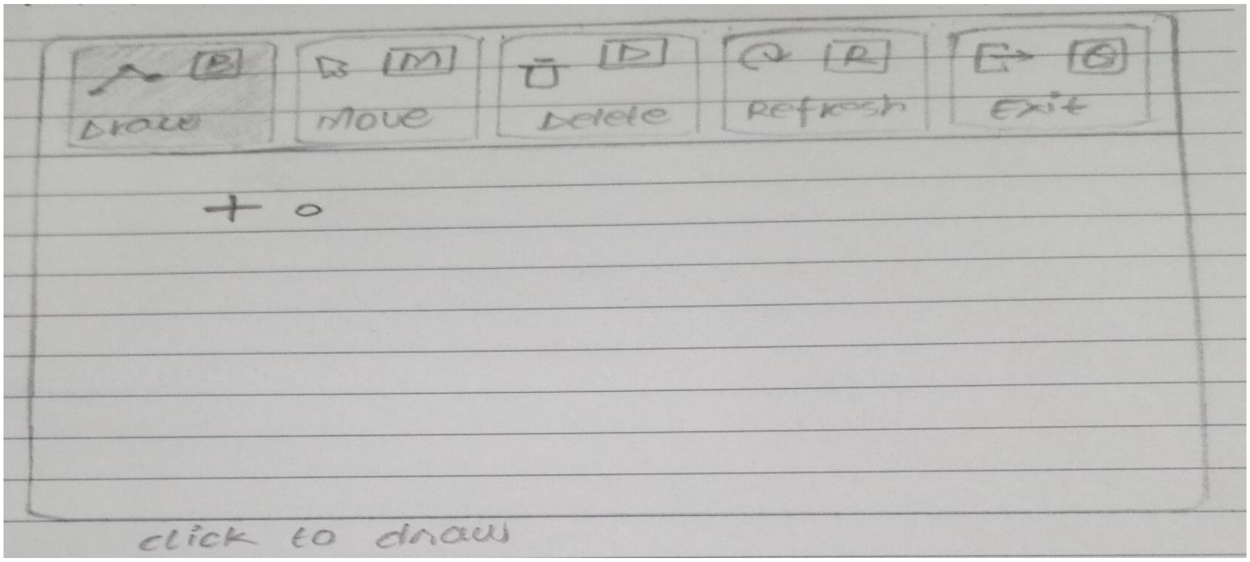


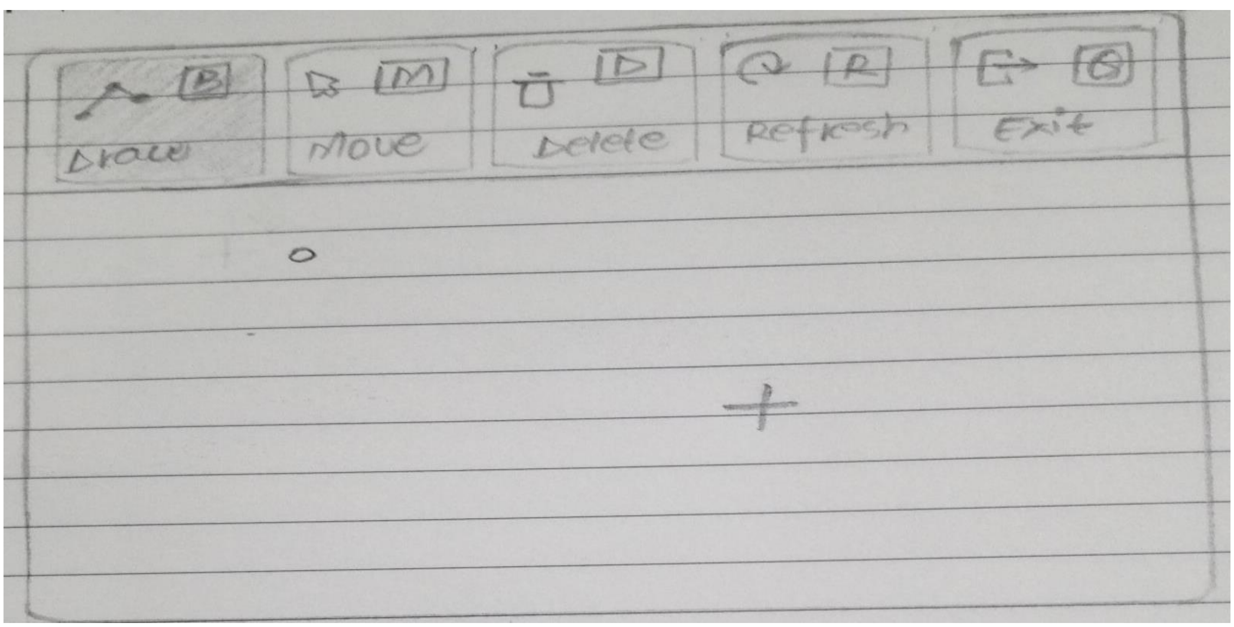
4. Mapping User Actions To Functions

- **Insertion**






B
Draw
M
Move
D
Delete
R
Refresh
Exit
click to draw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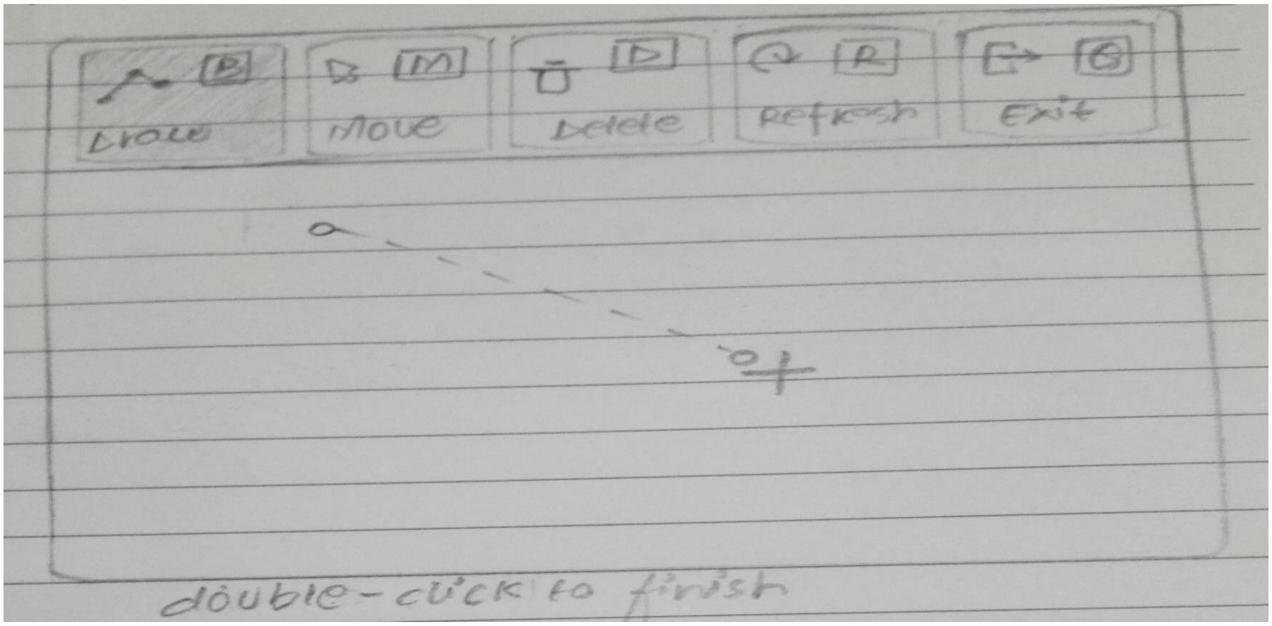
B
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Draw
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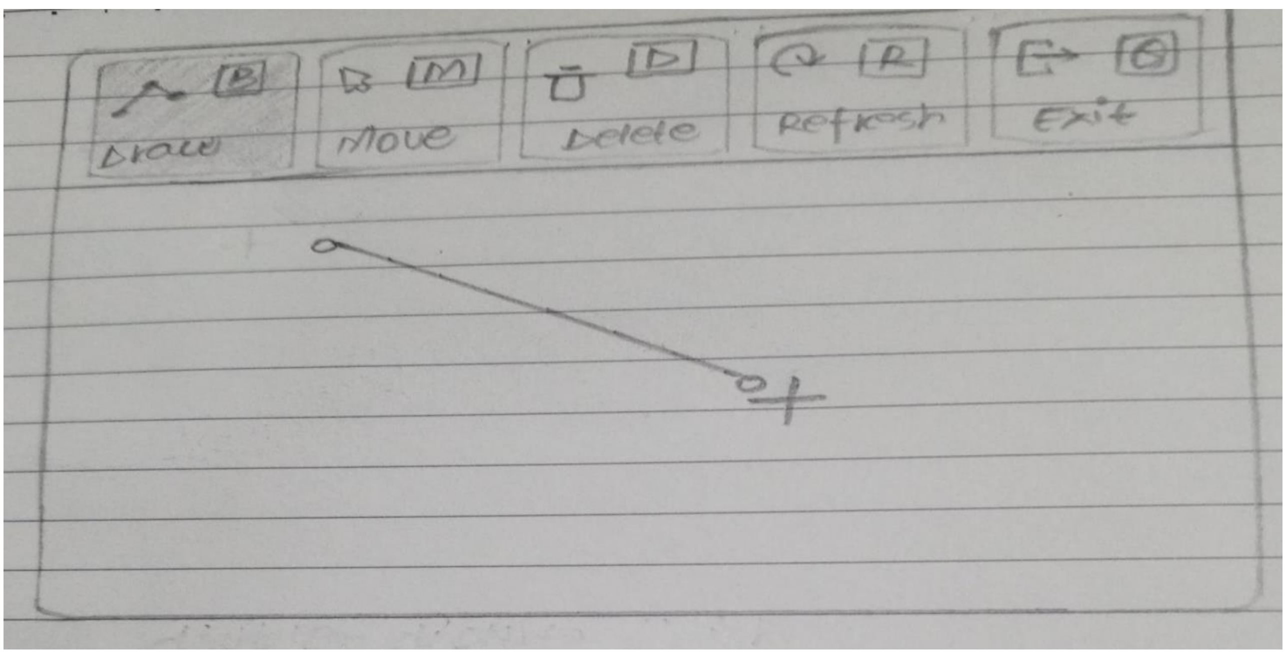
Refresh
Exit
double-click to finish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Draw
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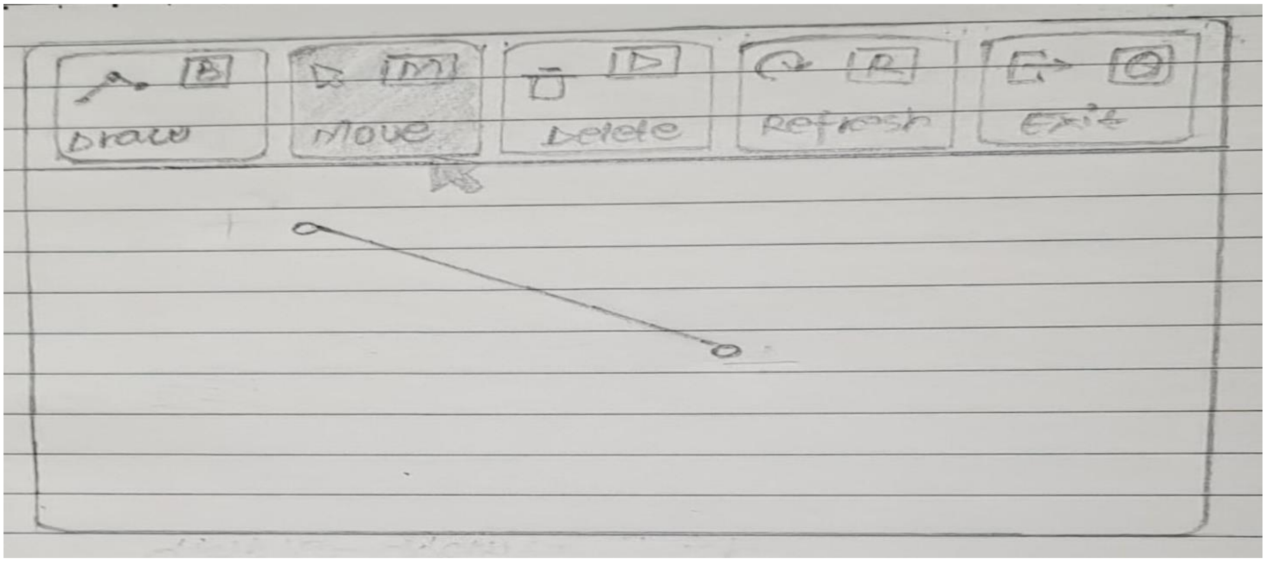
Refresh
Exit

- **Move**



B
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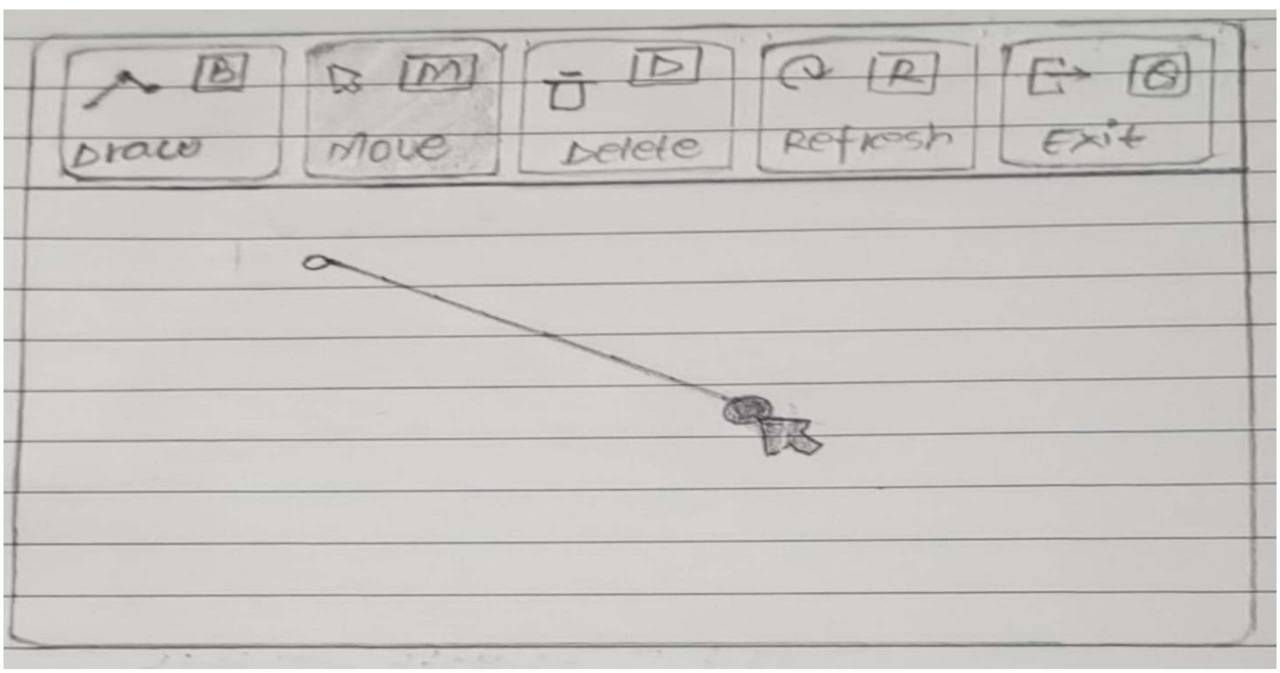
Draw
Move
Delete
Refresh
Exit



B
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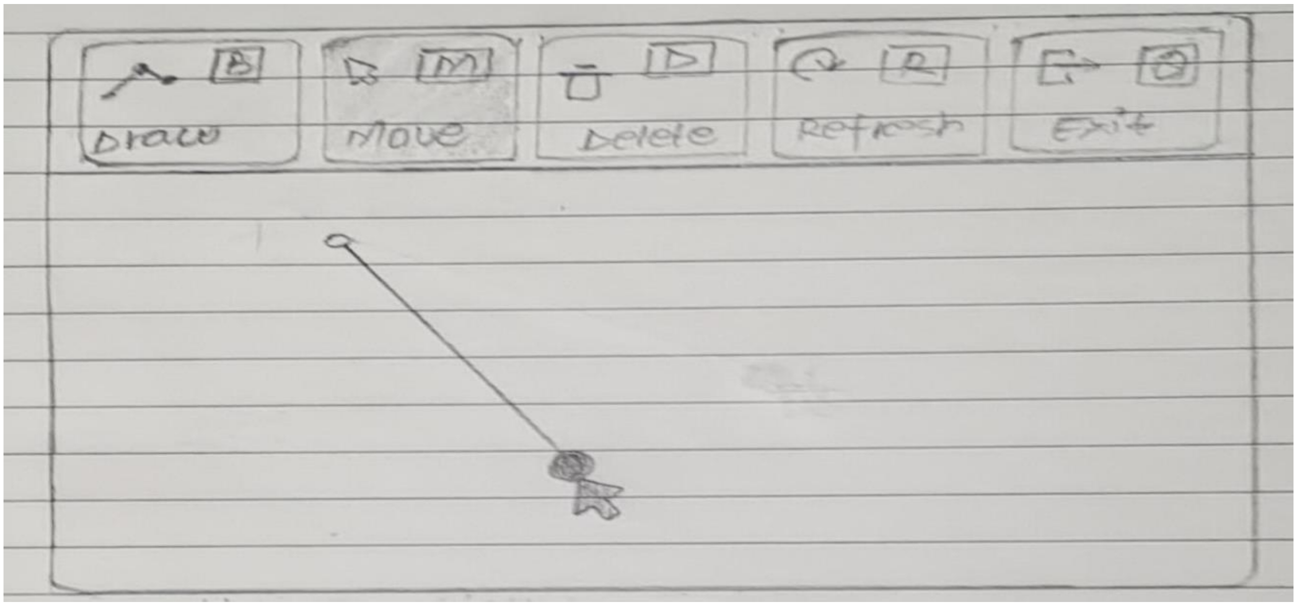
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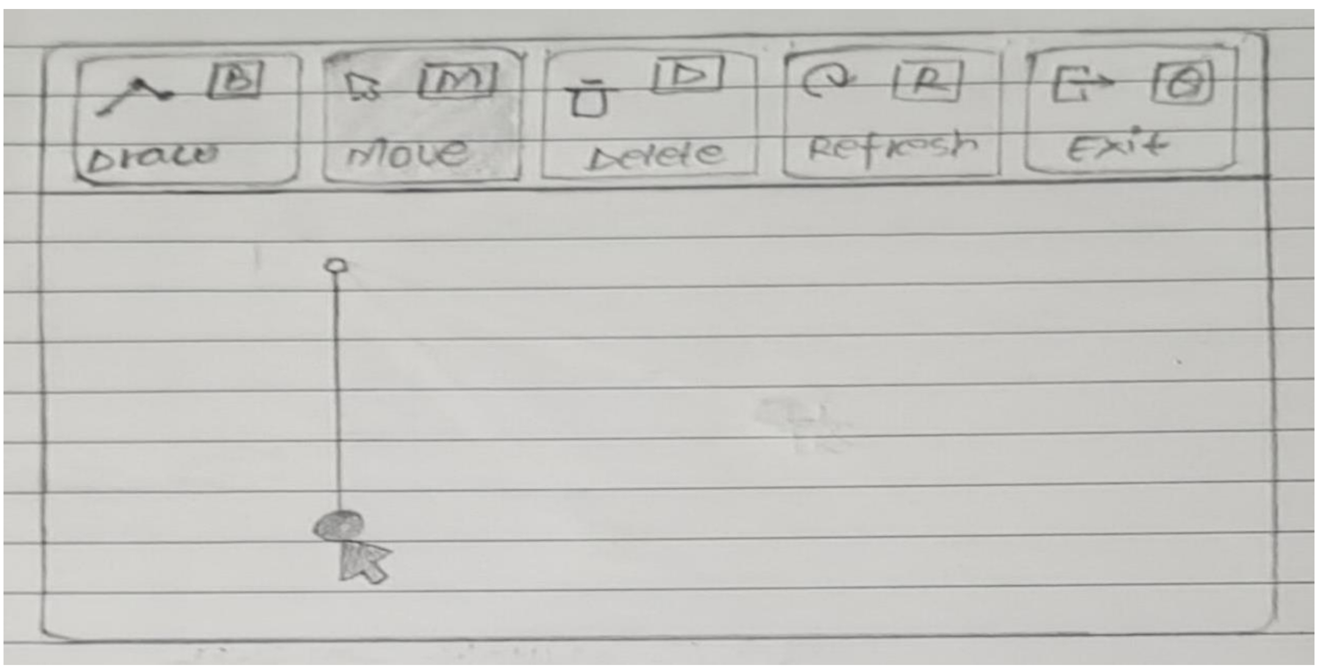
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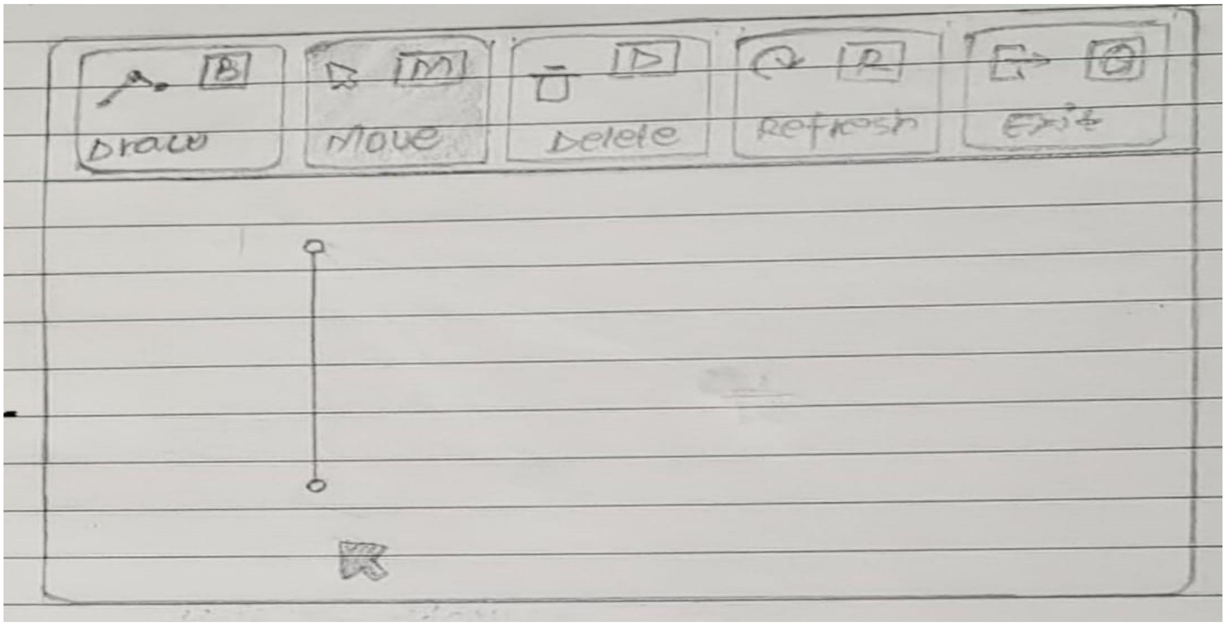
B
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Draw
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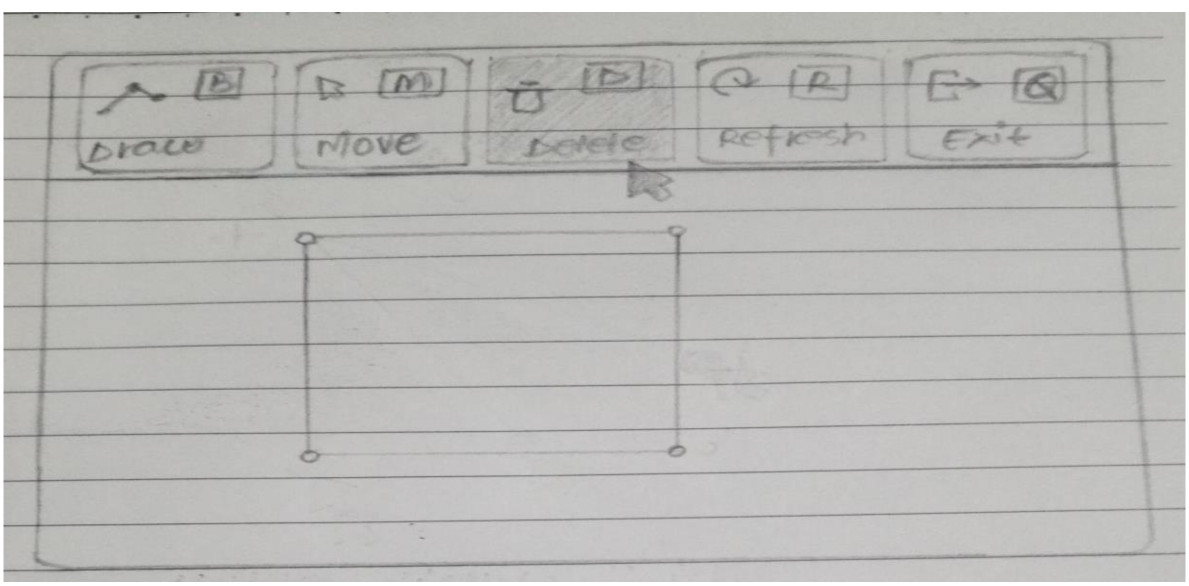
Refresh
Exit

- **DELETE**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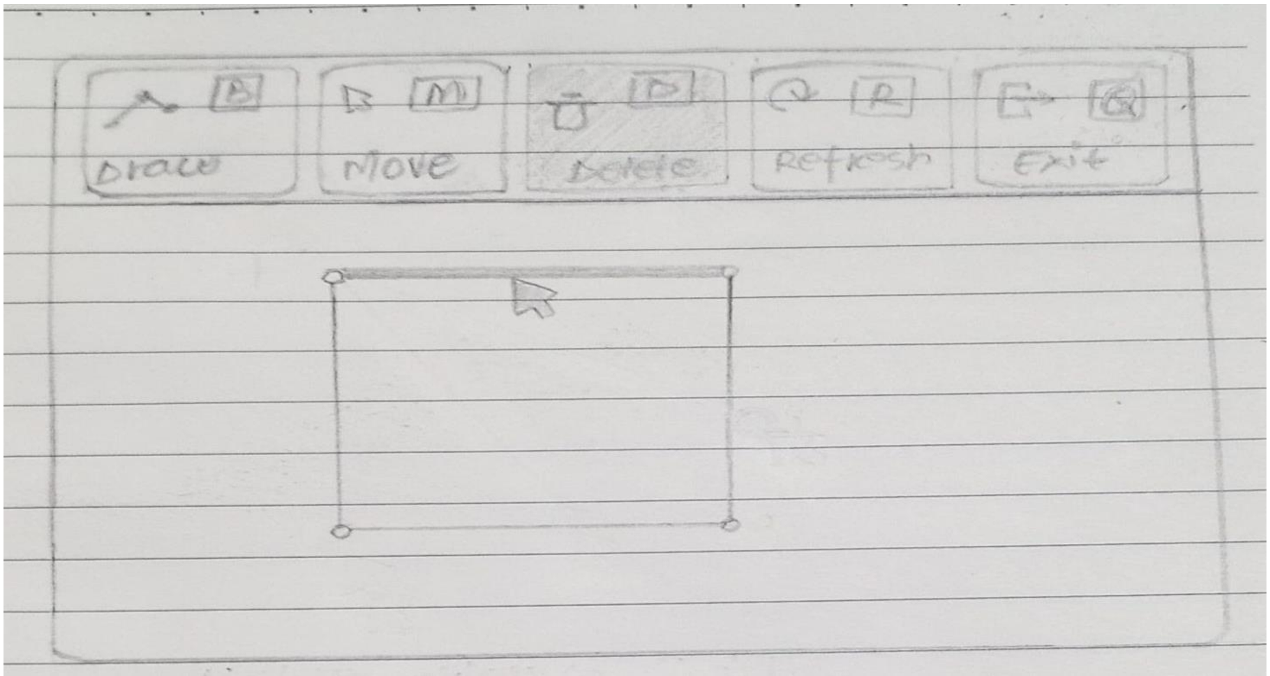
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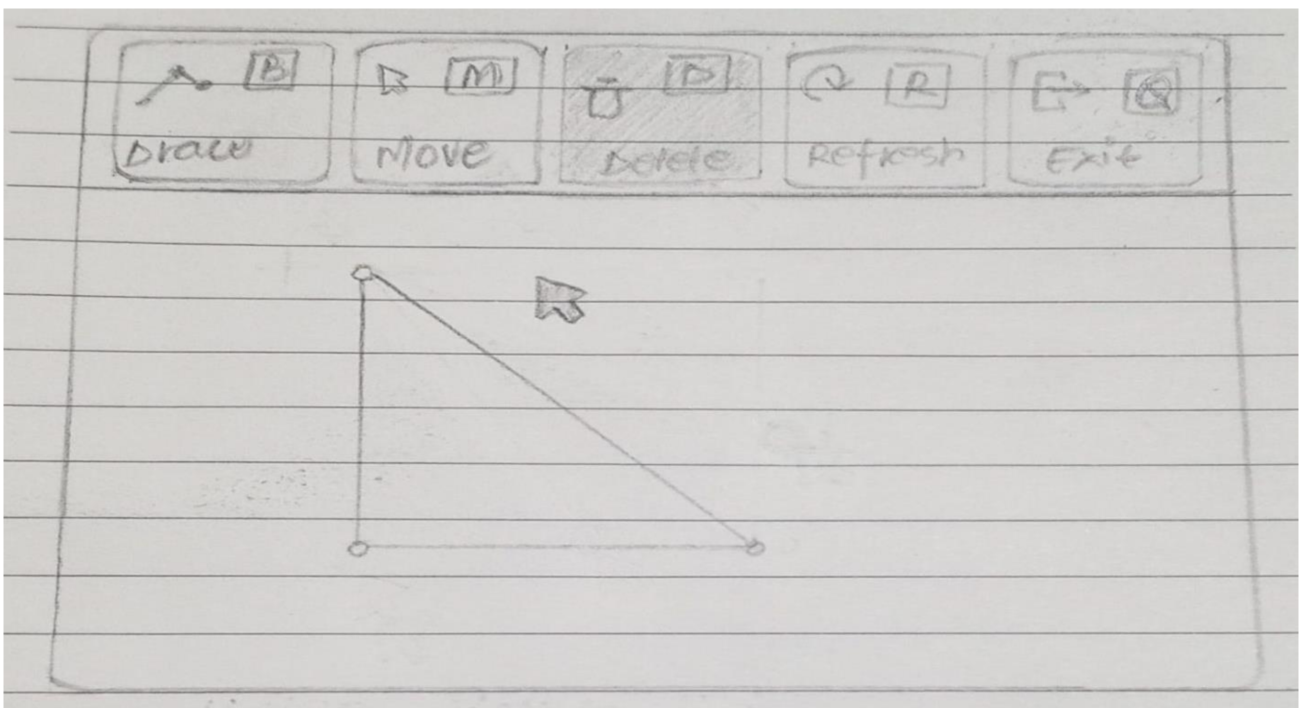
B
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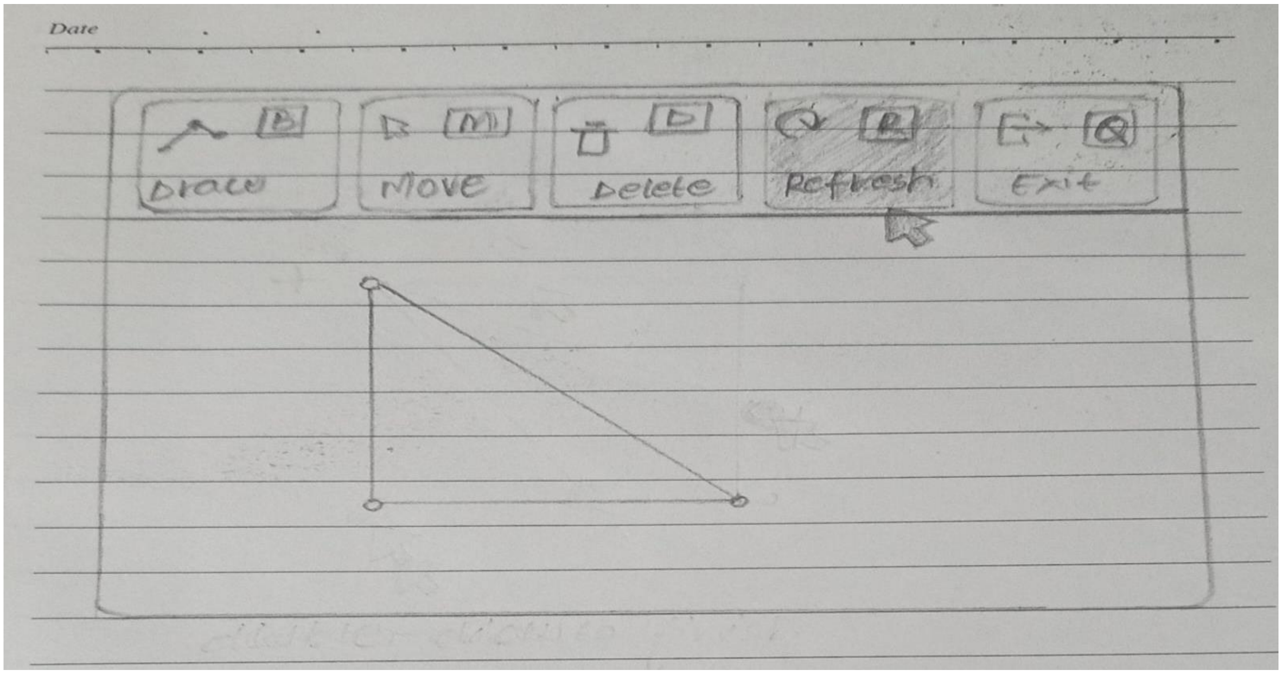
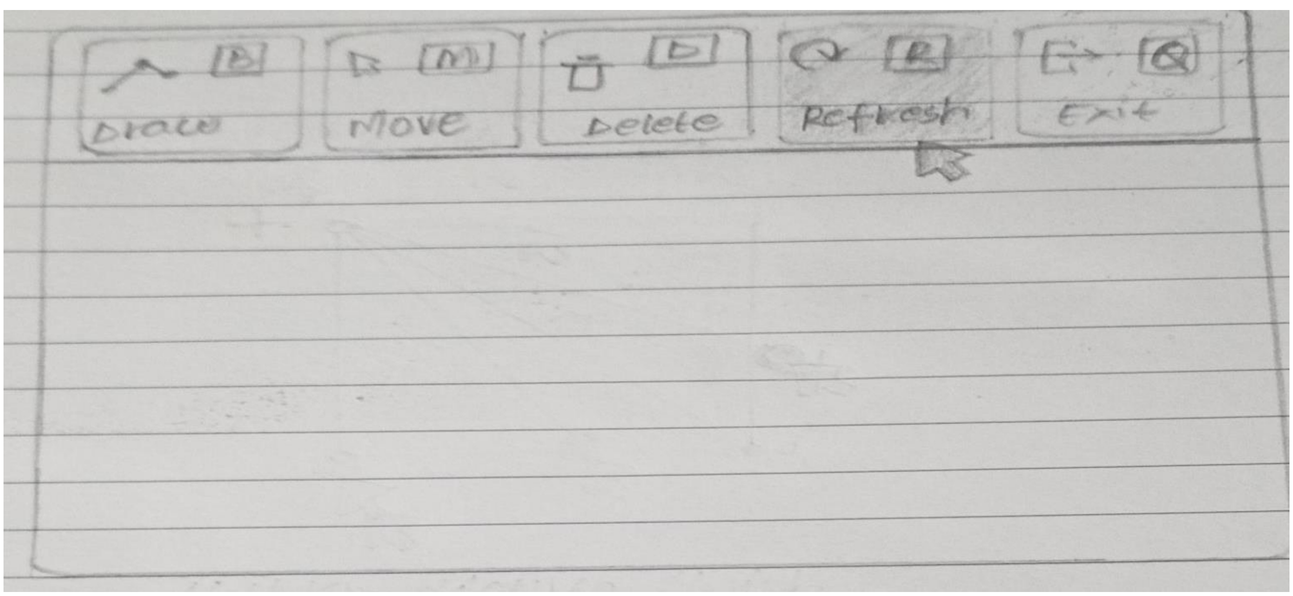
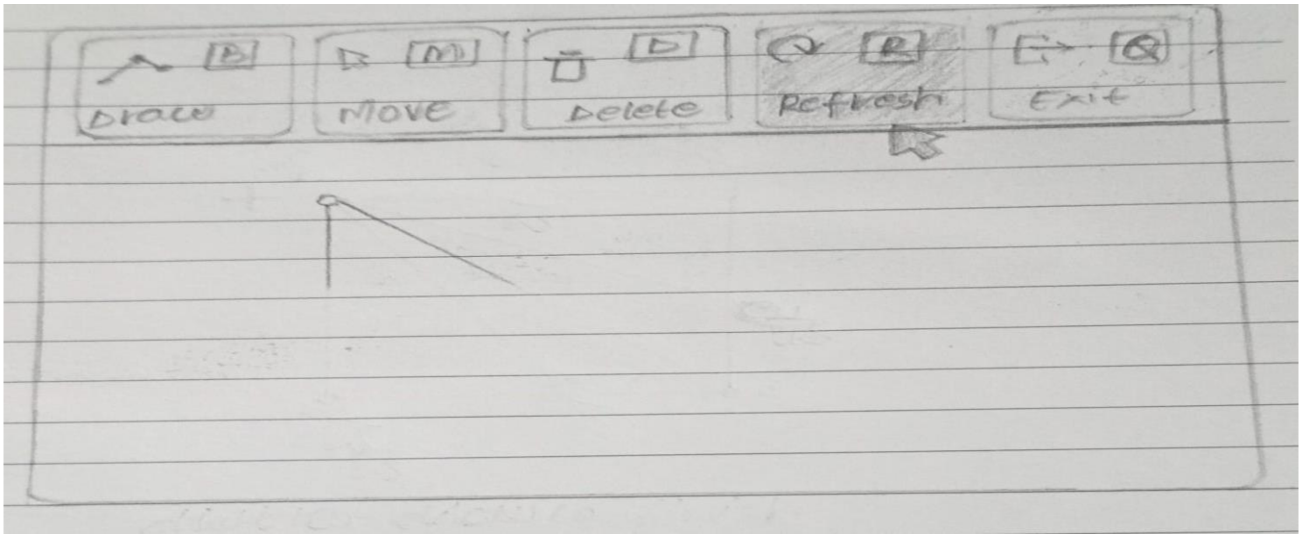
Draw
Move
Delete
Refresh
Exit

- **REFRESH**


Date
Draw
Move
Delete
Refresh
Exit


Draw
Move
Delete
Refresh
Exit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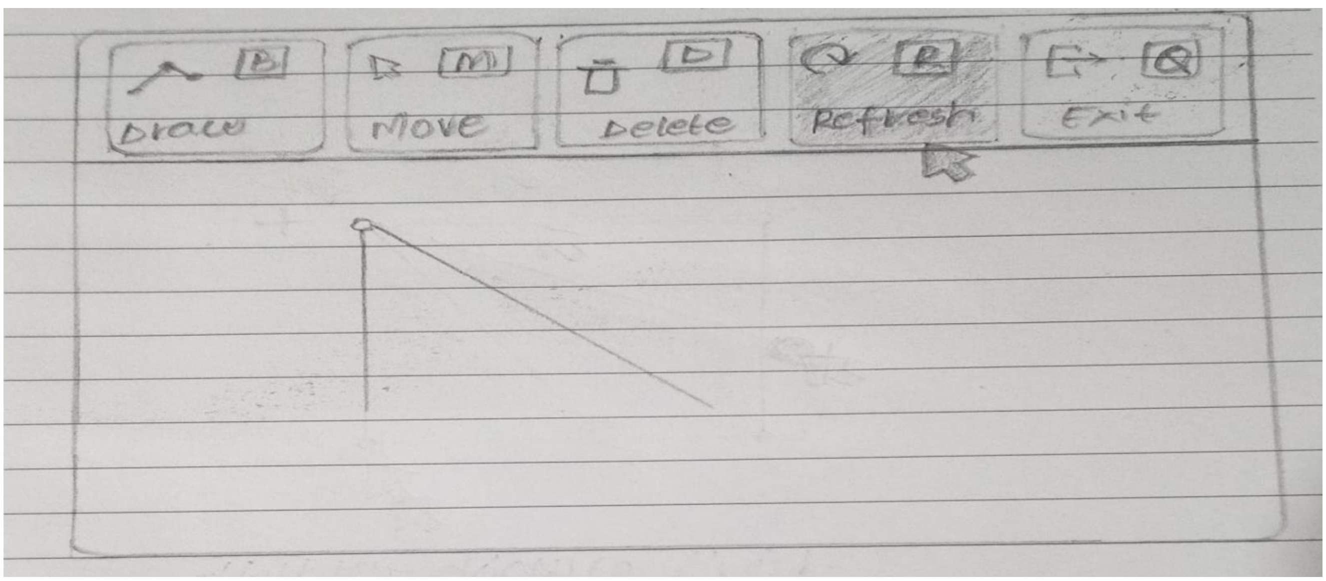
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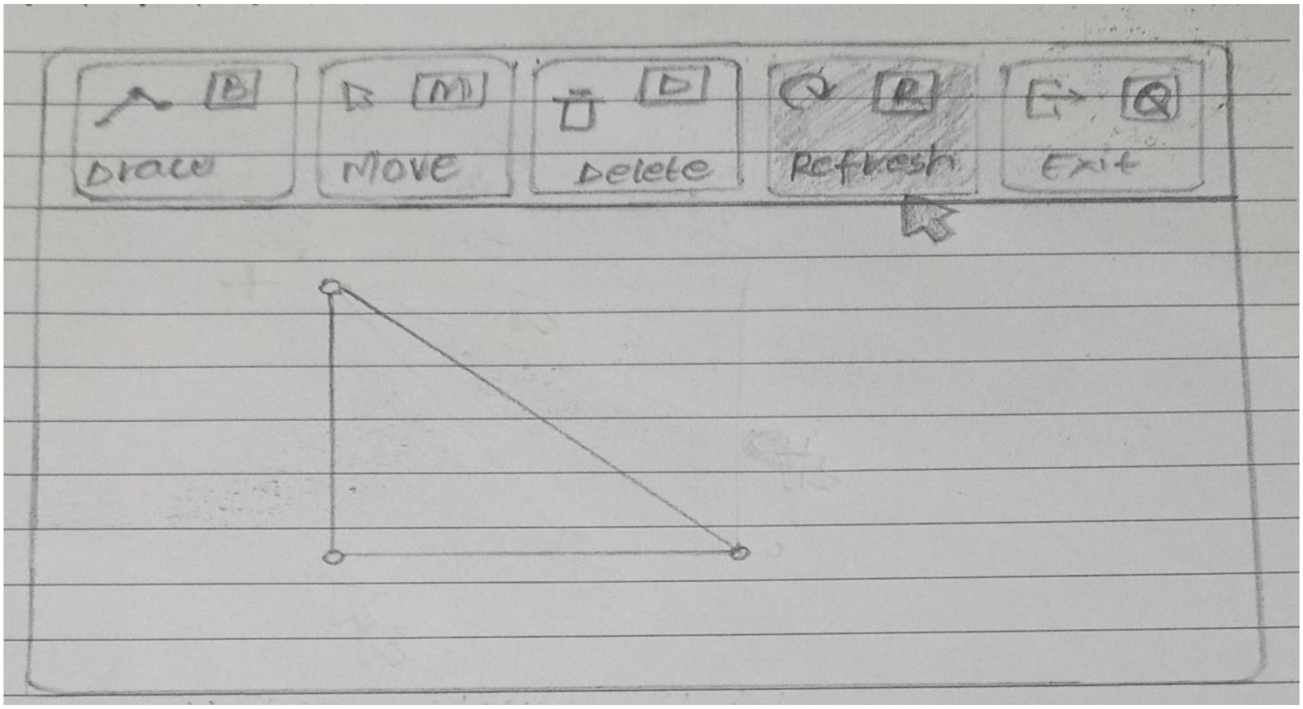
B
M
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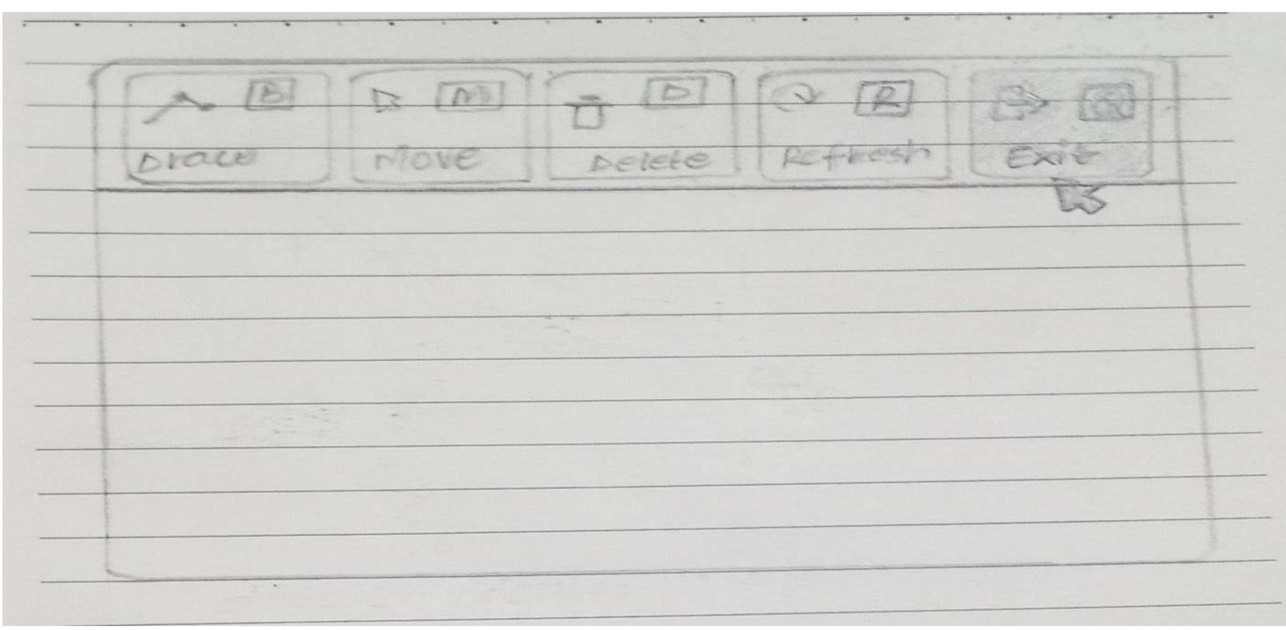
Q
Draw
Move
Delete
Refresh
Exit

- **EXIT**


B
M
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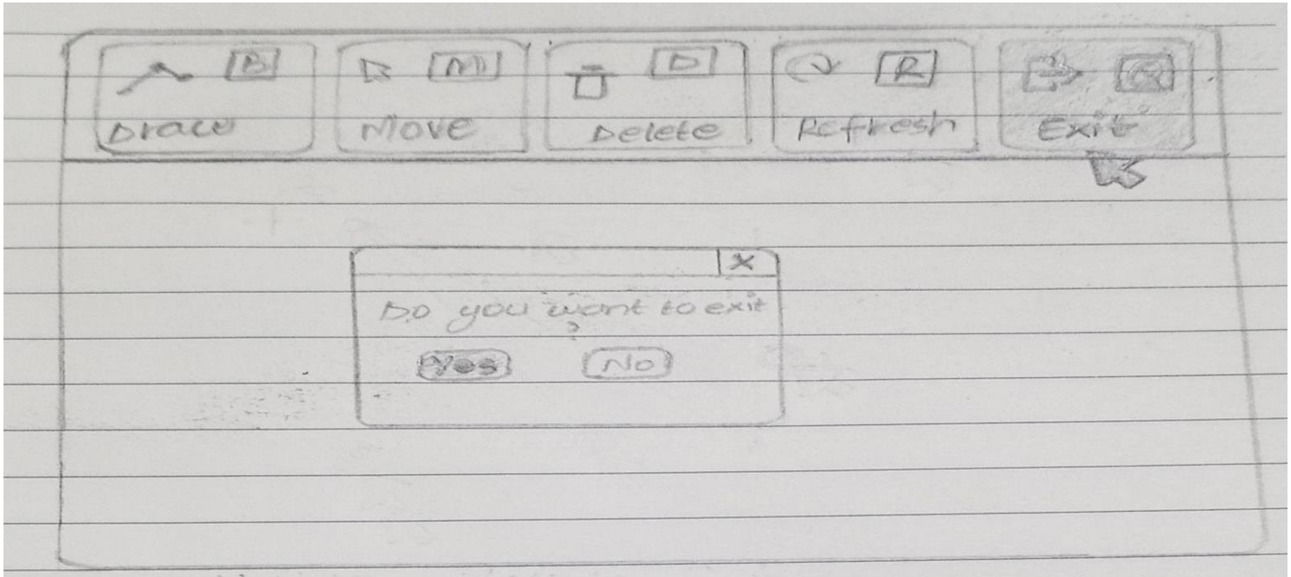
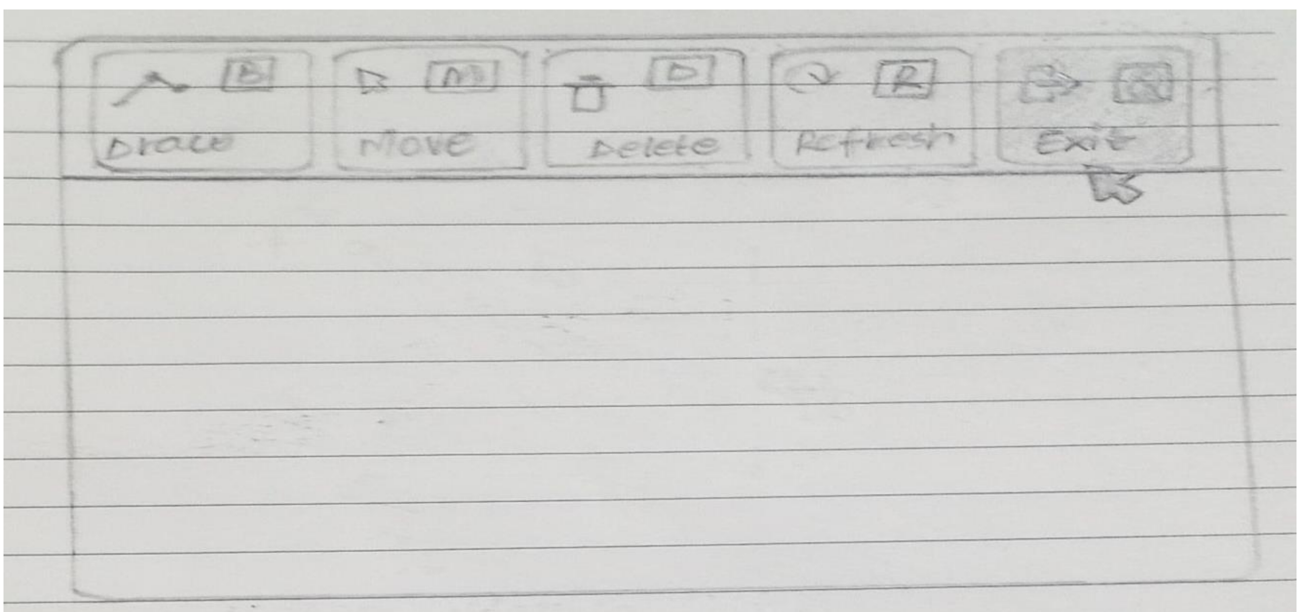
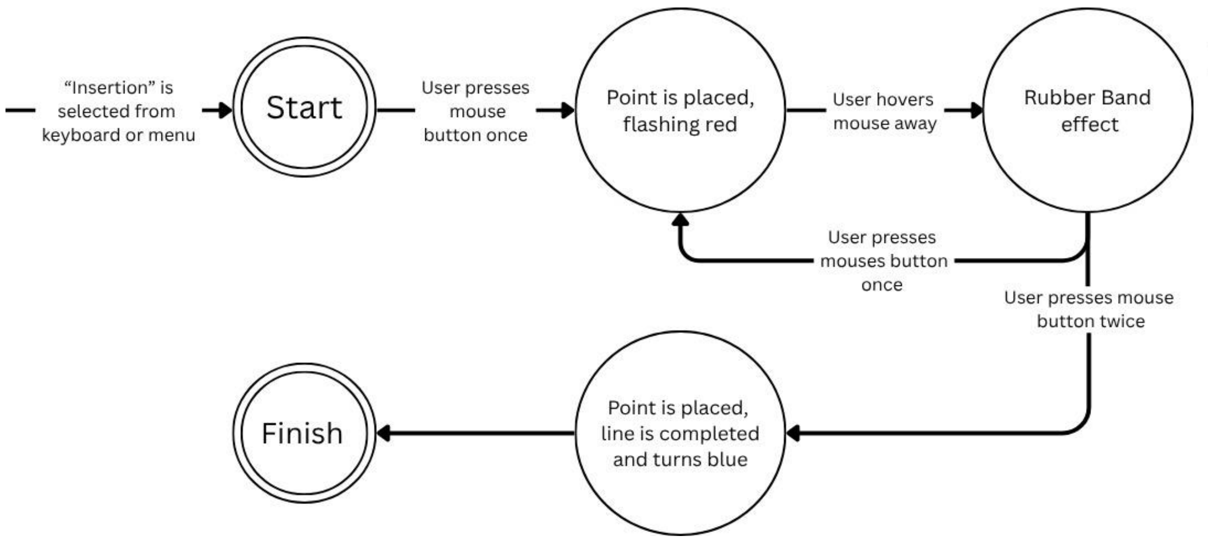
Q
Draw
Move
Delete
Refresh
Exit


B
Draw
M
Move
D
Delete
R
Refresh
Exit
Do you want to exit?
Yes
No

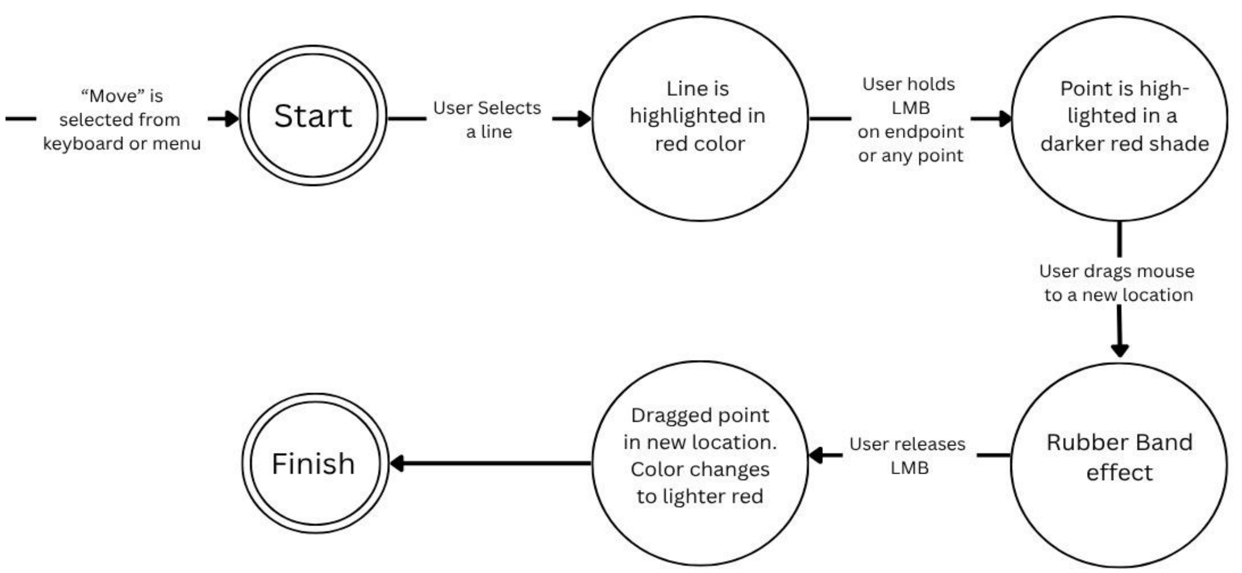

B
Draw
M
Move
D
Delete
R
Refresh
Exit

5.STATE TRANSITION NETWORK

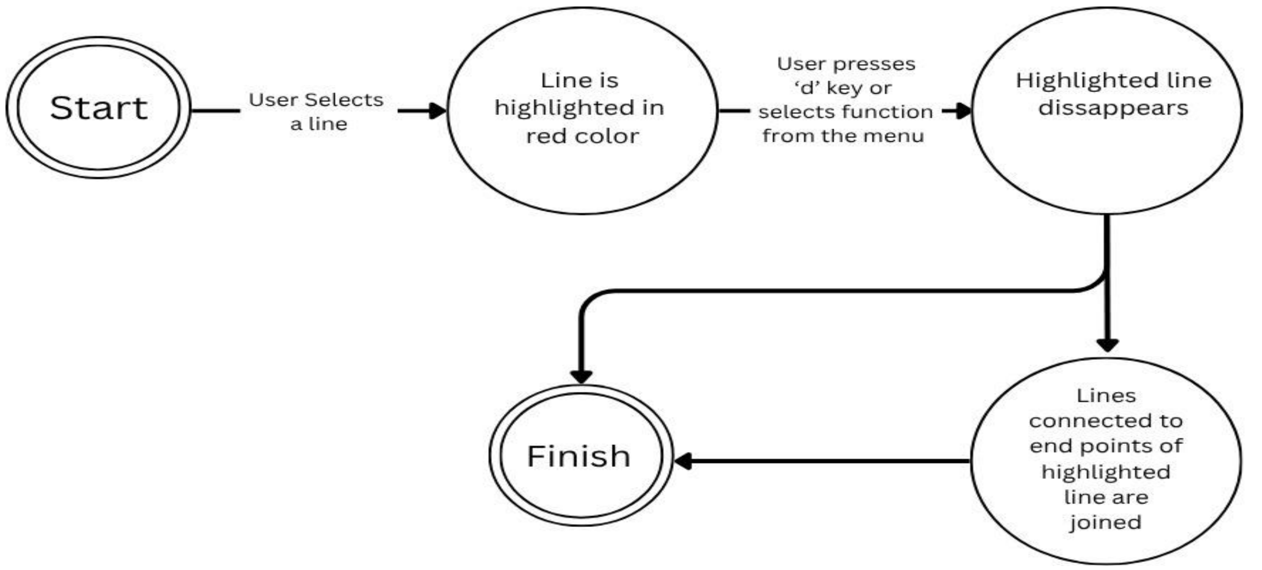
- **INSERTION**



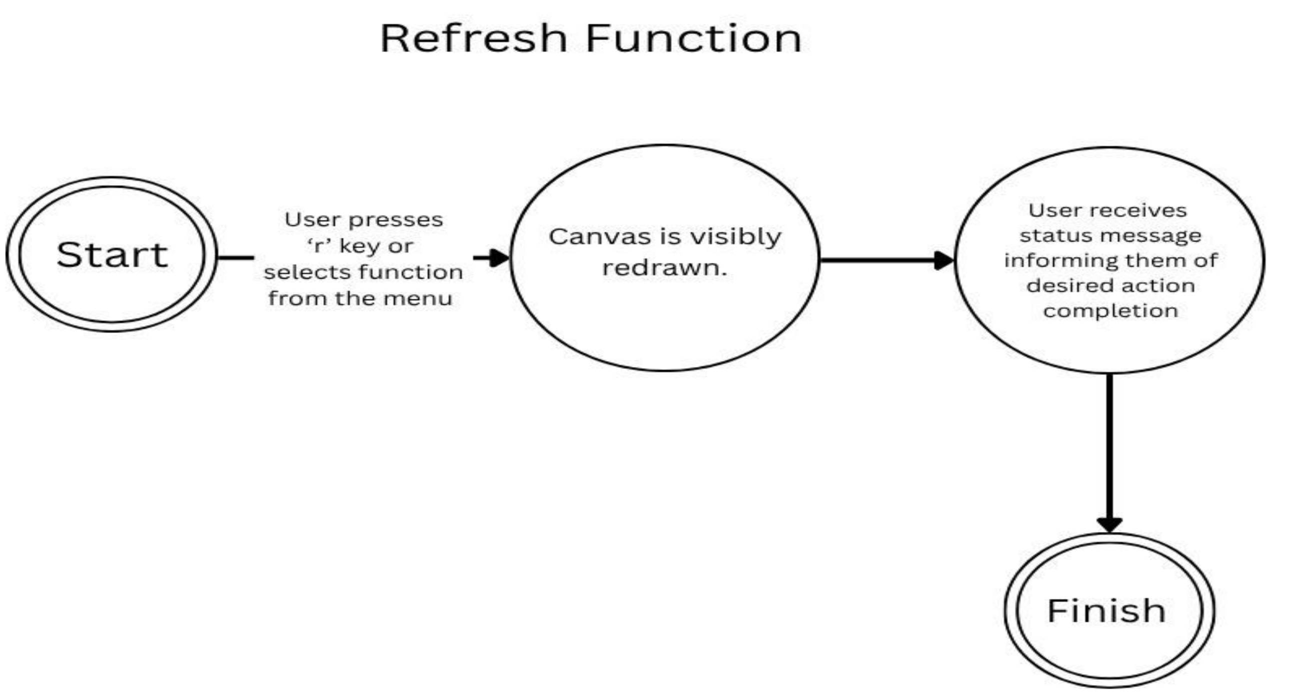
- **MOVE**



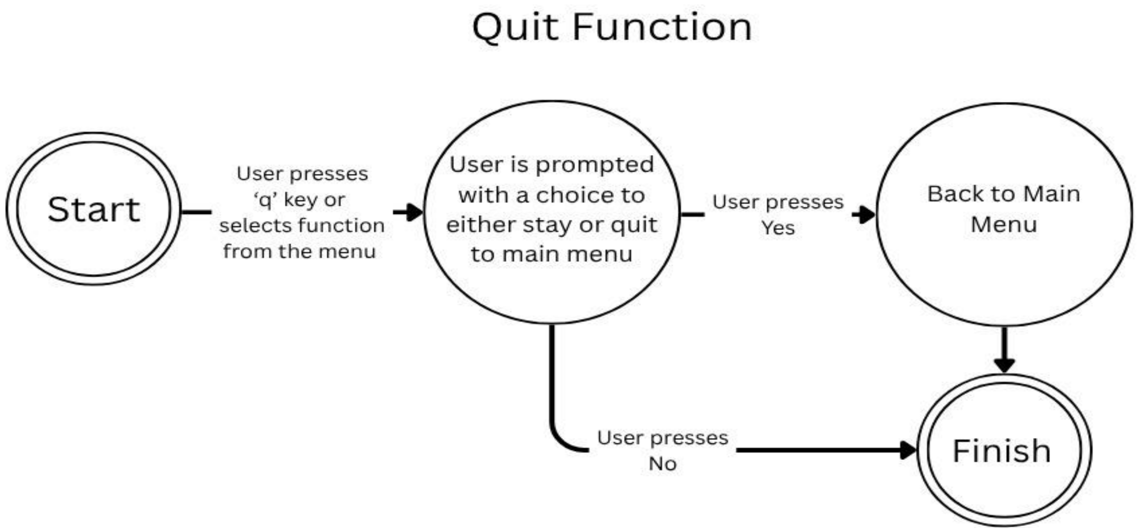
- **DELETE**



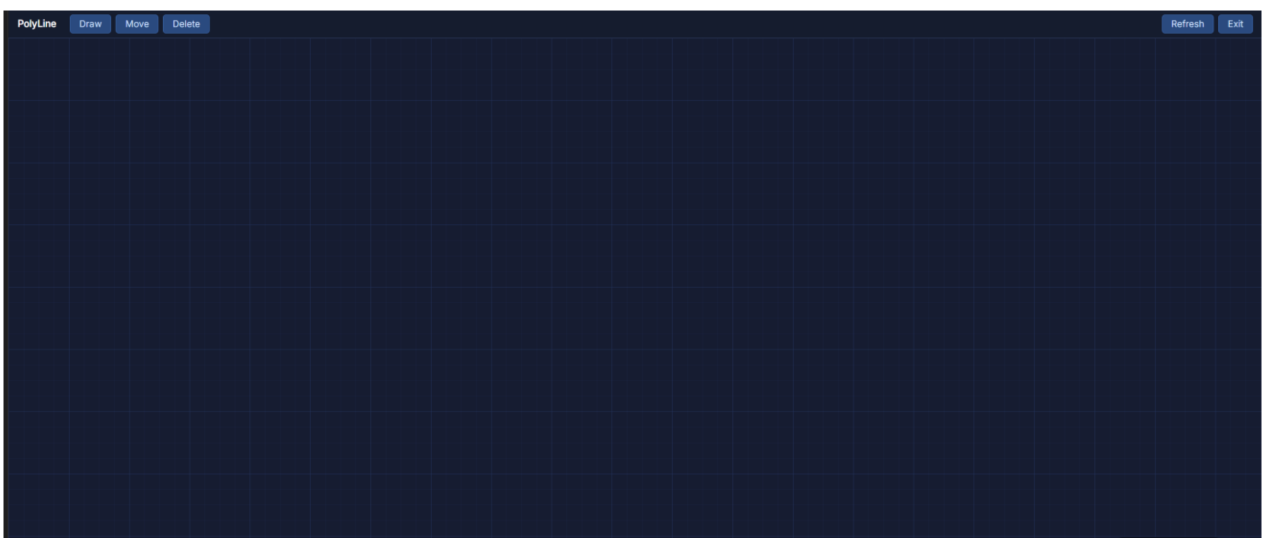
- **REFRESH**



- **QUIT**



7. Prototype Implementation (Version 1)



Prototype 1 Analysis – Polyline Editor

1. Introduction

This section evaluates Prototype 1 of the Polyline Editor to identify usability issues and design limitations. The analysis focuses on user interaction, system feedback, and overall usability based on HCI principles.

2. Evaluation Method

The prototype was evaluated by performing common user tasks such as drawing, moving, deleting, refreshing, and exiting. During this process, usability issues were identified by observing user interaction and comparing the system with basic HCI principles such as feedback, visibility, and consistency.

3. Identified Usability Issues

Lack of Mode Visibility

The system does not clearly indicate the current mode (Insert, Move, Delete), which may confuse users.

Insufficient User Feedback

The system does not provide feedback after actions, making it unclear whether actions are successful.

No Error Recovery Mechanism

There is no undo feature, so users cannot correct mistakes.

Limited Functionality

Only basic operations are supported, reducing usability.

Lack of Visual Highlighting

Selected points are not clearly highlighted, making interaction difficult.

No Data Persistence

There is no save/load option, causing loss of work.

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5. HCI Principle Violations

- Visibility of system status is weak
- Feedback is missing
- Error prevention is limited
- Error recovery is absent

6. Conclusion

- Prototype 1 provides basic functionality but lacks usability features and essential controls. The absence of feedback, error handling, and advanced features reduces system effectiveness. These limitations indicate the need for improvements in the next prototype.